

CAMILA FELIX DOS REIS

Computer Engineering Student | Backend Software Engineer | Industrial IoT | Digital Twins | Cyber-Physical Systems

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SUMMARY

Computer Engineering student (B.Eng., Dec 2028) with hands-on backend experience in industrial IoT and cyber-physical systems research. Develop and maintain Python/FastAPI services on Docker and Linux, integrating MQTT, OPC UA and Asset Administration Shell (Eclipse BaSyx) across distributed systems. Independently own the end-to-end software development lifecycle of research projects, from software architecture and system design to implementation, integration, testing, validation and technical documentation.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, JavaScript, TypeScript, SQL

Backend & APIs: FastAPI, REST APIs, API Design, Distributed Systems, Asynchronous Communication, System Design

Infrastructure: Docker, Linux, Git, GitHub, GitLab

Cloud & Databases: AWS Lambda, AWS S3, AWS Rekognition, PostgreSQL, TimescaleDB, MongoDB

Industrial IoT: MQTT, OPC UA, Asset Administration Shell (AAS), Eclipse BaSyx, RAMI 4.0, Digital Twins

Engineering Practices: Software Architecture, Modular Design, Automated Testing, Debugging, Git Workflow, UML, PlantUML, LaTeX (Overleaf)

PROFESSIONAL EXPERIENCE

CyberLabs / UEA — Software Engineer (R&D)

Aug 2025 – Present

Manaus, Brazil • Cyber-Physical Systems research group

- Develop backend services in Python and FastAPI for cyber-physical systems research, containerized with Docker on Linux.
- Implement new features and integrate services across MQTT, OPC UA and Asset Administration Shell (Eclipse BaSyx).
- Contribute to technical decisions while evolving a legacy distributed platform, refactoring modules toward maintainable code.
- Debug and validate software against real equipment data, troubleshooting integration and time-series historization issues.
- Write and version technical documentation (LaTeX/Overleaf, PlantUML, UML); manage code and docs in Git, GitHub and GitLab.

Mach Force (Shell Eco-marathon) — Hardware & Software Technical Lead

Feb 2026 – Present

- Lead hardware/software integration for the team's Shell Eco-marathon prototype, coordinating telemetry architecture, backend integration and real-time monitoring.

TOTVS Norte — Technical Support Analyst

Sep 2024 – Jun 2025

- Automated recurring SQL reports, analyzed relational databases and resolved production issues in ERP environments.

INDEPENDENT RESEARCH PROJECTS

Vehicle Energy Digital Twin — Sole Engineer

Python, FastAPI, CARLA, MQTT, OPC UA, BaSyx

- Architected and built an end-to-end Digital Twin of an electric prototype vehicle's energy system, modeling the asset with the Asset Administration Shell standard on Eclipse BaSyx.
- Designed and implemented the distributed communication architecture, integrating CARLA, MQTT, OPC UA, PostgreSQL/TimescaleDB and FastAPI REST APIs for real-time monitoring.
- Designed a layered software architecture following separation of concerns and modular design principles.
- Containerized the full stack with Docker; engineered automated tests, validated energy results and authored the technical documentation.

CloudEdge Medicinal Plants — Sole Engineer

AWS Lambda, AWS S3, AWS Rekognition

- Designed an offline-first cloud-edge architecture for medicinal plant identification: edge captures sync to AWS S3 and run inference through AWS Rekognition, orchestrated by AWS Lambda.

Offline Cyber-Physical System for Aquaculture — Sole Engineer

MQTT, OPC UA, Machine Learning

- Architected an offline-first cyber-physical system for water-quality monitoring, integrating MQTT and OPC UA communication with Random Forest prediction and Genetic Algorithm optimization; implemented the backend and validated results.

EDUCATION

Faculdade Matias Machline — Bachelor of Computer Engineering

Jan 2024 – Dec 2028 (expected)

Manaus, Amazonas, Brazil • **Languages:** Portuguese (Native), English (B1 – Intermediate)

AWARDS & CERTIFICATIONS

Award: 2nd Place — Faculdade Matias Machline Engineering Fair

Certifications: Containers & Docker • Industry 4.0 (RAMI 4.0 / AAS) • Artificial Intelligence • REST APIs with C#/.NET